

0620 • PAPER 4

TOPIC

09

Metals

Questions only • Original examination pages retained

20

QUESTIONS

253

TOTAL MARKS

1

SECTIONS



USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

Quick navigation and progress overview

1/1

01

Properties, extraction, reactivity and corrosion of metals

20 questions • 253 marks

9.1 Properties, extraction, reactivity and corrosion of metals

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	2020 Oct/Nov Paper 41 Question 1	9	☐	☐	☐
02	2020 Oct/Nov Paper 41 Question 2	9	☐	☐	☐
03	2020 Oct/Nov Paper 42 Question 3	12	☐	☐	☐
04	2020 Oct/Nov Paper 43 Question 1	6	☐	☐	☐
05	2021 May/June Paper 41 Question 6	13	☐	☐	☐
06	2021 May/June Paper 43 Question 6	13	☐	☐	☐
07	2021 Oct/Nov Paper 41 Question 1	18	☐	☐	☐
08	2022 May/June Paper 41 Question 1	9	☐	☐	☐
09	2022 May/June Paper 42 Question 1	8	☐	☐	☐
10	2023 May/June Paper 42 Question 1	6	☐	☐	☐
11	2023 Oct/Nov Paper 41 Question 3	12	☐	☐	☐
12	2023 Oct/Nov Paper 42 Question 3	17	☐	☐	☐
13	2023 Oct/Nov Paper 43 Question 5	18	☐	☐	☐
14	2024 Feb/March Paper 42 Question 1	19	☐	☐	☐

9.1 Properties, extraction, reactivity and corrosion of metals

#	SOURCE	MARKS	DONE	MARK	REVIEW
15	2024 May/June Paper 41 Question 6	14	☐	☐	☐
16	2024 May/June Paper 42 Question 2	14	☐	☐	☐
17	2024 May/June Paper 43 Question 6	13	☐	☐	☐
18	2024 Oct/Nov Paper 43 Question 1	9	☐	☐	☐
19	2025 May/June Paper 41 Question 3	15	☐	☐	☐
20	2025 Oct/Nov Paper 42 Question 2	19	☐	☐	☐

- 1 (a) This question is about elements.

aluminium
carbon
iron
hydrogen
oxygen
silicon
sodium
sulfur

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

- (i) Name the element that can be used as a fuel.
..... [1]
- (ii) Name the element that forms an oxide with a similar structure to diamond.
..... [1]
- (iii) Name the element that forms an amphoteric oxide.
..... [1]
- (iv) Name the element that has oxidation states of +2 and +3.
..... [1]
- (v) Name the element extracted from bauxite.
..... [1]
- (vi) Name the element that has atoms with the electronic structure 2,6.
..... [1]

(b) Iron rusts when it is in contact with oxygen and water.

(i) Explain how sacrificial protection prevents rusting.

.....

.....

.....

..... [2]

(ii) State one **other** method of rust prevention.

..... [1]

[Total: 9]

2 Zinc is extracted from an ore containing zinc sulfide.

(a) State the name of this zinc ore.

..... [1]

(b) This ore is converted to zinc oxide, ZnO.

Zinc oxide is then reacted with carbon.

(i) Write a chemical equation for the reaction of zinc oxide with carbon.

..... [1]

(ii) State what type of chemical change happens to the zinc in zinc oxide in this reaction.

Explain your answer.

chemical change

explanation

.....

.....

[2]

(iii) Explain why aluminium is **not** extracted from aluminium oxide by heating with carbon.

.....

..... [1]

(iv) Suggest an alternative method for the extraction of zinc from zinc oxide.

..... [1]

(c) Brass is an alloy of zinc.

Explain, in terms of particles, why brass is harder than pure zinc.

.....

.....

.....

.....

..... [3]

[Total: 9]

- 3 Group I metals are very reactive. Transition elements are also metals but are less reactive than Group I metals.

(a) State **two** physical properties of Group I metals which are similar to those of transition metals.

1

2 [2]

(b) Describe **two** ways in which the physical properties of Group I metals are different from those of transition metals.

1

.....

2

..... [2]

(c) When Group I metals are added to water they fizz and an alkaline solution forms.

(i) Name the gas given off.

..... [1]

(ii) Identify the ion present in the solution which makes the solution alkaline.

..... [1]

(iii) Write the chemical equation for the reaction between sodium and water.

..... [2]

(d) When the transition element iron is added to water the iron rusts.

When an iron object is coated with a layer of zinc, rusting is prevented.

(i) Name this process of coating iron objects with a layer of zinc.

..... [1]

(ii) Explain how completely coating an iron object with a layer of zinc prevents rusting.

..... [1]

(iii) Rusting of iron ships can be prevented by attaching zinc blocks to the hull of the ship.

Explain how this prevents rusting.

.....

..... [2]

[Total: 12]

- 1 The names of nine substances are shown.

aluminium oxide
ammonia
carbon monoxide
anhydrous cobalt(II) chloride
hydrated copper(II) sulfate
iron(III) oxide
nitrogen dioxide
silver
steel

Answer the following questions using these substances. Each substance may be used once, more than once or not at all.

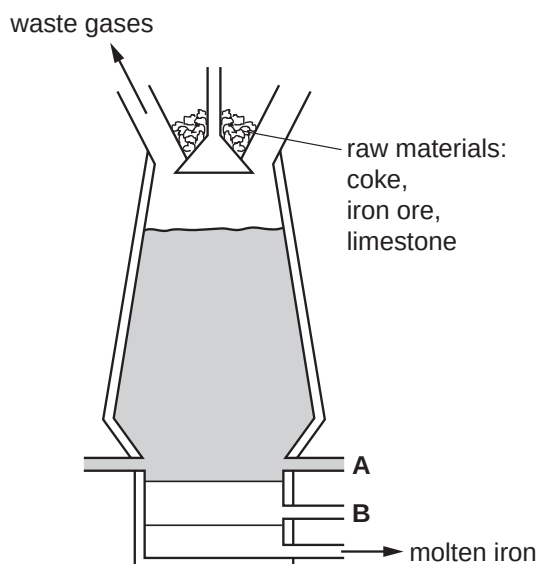
Name the substance that is:

- (a) the main constituent of hematite [1]
- (b) a gas produced in car engines which causes acid rain [1]
- (c) an alkaline gas [1]
- (d) an element [1]
- (e) a gas formed by the incomplete combustion of fossil fuels [1]
- (f) used to test for the presence of water. [1]

[Total: 6]

6 This question is about iron.

(a) Iron is extracted from its main ore in a blast furnace.



(i) Name the main ore of iron used in the blast furnace.

..... [1]

(ii) Name the substance that enters the blast furnace at **A**.

..... [1]

(iii) Name the substance that leaves the blast furnace at **B**.

..... [1]

(iv) Give **two** reasons for using coke in the blast furnace.

1

2

[2]

(b) Another ore of iron is iron pyrites, FeS_2 . Iron pyrites contains the positive ion, Fe^{2+} .

Deduce the formula of the negative ion in FeS_2 .

..... [1]

(c) Iron is a transition element.

A list of properties of iron is shown.

- Iron is a good conductor of electricity.
- Iron forms soluble salts.
- Iron forms coloured compounds.
- Iron has variable oxidation states.
- Iron acts as a catalyst.
- Iron forms a basic oxide.

(i) Give **two** properties from the list in which iron differs from Group I elements.

1

2 [2]

(ii) Give **two** properties from the list in which iron is similar to Group I elements.

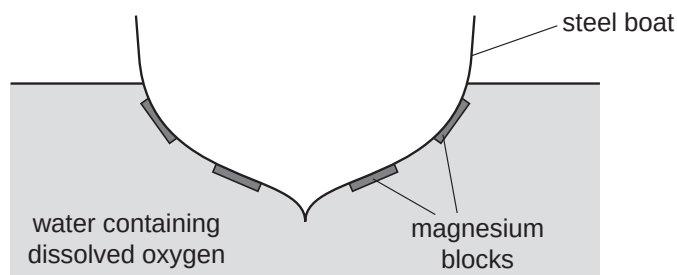
1

2 [2]

(d) Steel consists mainly of iron.

Iron forms rust when it reacts with water and oxygen.

Magnesium blocks can be attached to the bottom of steel boats. The magnesium does not completely cover the steel.



(i) Explain how the magnesium blocks prevent iron from rusting.

.....
.....
.....
..... [2]

- (ii) Explain why replacing the magnesium blocks with copper blocks will **not** prevent the bottom of the boat from rusting.

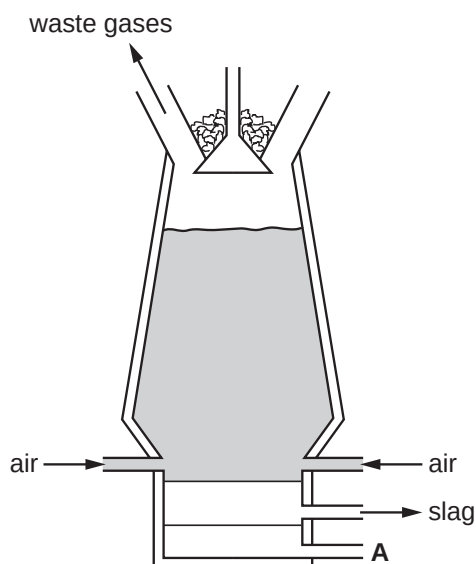
.....

..... [1]

[Total: 13]

6 This question is about metals.

(a) Iron is extracted from its main ore in a blast furnace.



(i) Coke and iron ore are added at the top of the blast furnace.

Name one **other** substance that is added at the top of the blast furnace.

..... [1]

(ii) Name the substance that leaves the blast furnace at **A**.

..... [1]

(iii) Iron ore is mainly iron(III) oxide, Fe_2O_3 .

Name a substance that reduces iron(III) oxide to iron in the blast furnace.

..... [1]

(iv) Temperatures inside a blast furnace can reach 2000°C .

Name **two** substances that react together, in the blast furnace, to produce this high temperature.

..... [1]

(v) Name **two** waste gases that leave the blast furnace.

1

2

[2]

(b) Zinc is extracted from zinc blende.

(i) Name the main zinc compound that is present in zinc blende.

..... [1]

(ii) When zinc is extracted, it is formed as a gas.

The gaseous zinc is then converted into molten zinc.

State the name of this physical change.

..... [1]

(c) Name the alloy that contains zinc and copper only.

..... [1]

(d) Copper has the following properties.

- It has a high melting point.
- It has a high density.
- It is a good conductor of electricity.
- It has variable oxidation states.
- It forms a basic oxide.
- It forms soluble salts.

(i) Give **two** properties from the list in which copper differs from Group I elements.

1

2 [2]

(ii) Give **two** properties from the list in which copper is similar to Group I elements.

1

2 [2]

[Total: 13]

- 1 Some elements are shown in the order they appear in the reactivity series. The most reactive element is at the top.

sodium
calcium
magnesium
aluminium
zinc
iron
hydrogen
copper

- (a) Answer the questions using the list of elements. Each element may be used once, more than once or not at all.

Identify:

- (i) a non-metal

..... [1]

- (ii) a metal which is stored under oil

..... [1]

- (iii) the main component of steel

..... [1]

- (iv) a metal with three electrons in the outer shell of its atoms

..... [1]

- (v) a metal found in brass

..... [1]

- (vi) a metal that forms chlorides of the type XCl_2 and XCl_3 .

..... [1]

- (b) Name the main ores of:

- (i) zinc [1]

- (ii) aluminium. [1]

- (c) In an experiment, a sample of aluminium appeared less reactive than expected.

Explain why.

.....

..... [1]

(d) Name **two** metals from the list which are extracted by reduction of their ores using carbon.

1

2 [2]

(e) When zinc granules are added to aqueous copper(II) sulfate, a reaction occurs. During the reaction, a red-pink solid is formed and the solution becomes colourless.

(i) Name the red-pink solid.

..... [1]

(ii) Name the colourless solution.

..... [1]

(iii) Explain, in terms of particles, why the rate of this reaction increases when the temperature is increased.

.....

.....

.....

.....

.....

..... [3]

(iv) Suggest two **other** ways of increasing the rate of this reaction.

1

2 [2]

[Total: 18]

1 A list of substances is shown.

aluminium oxide	carbon dioxide	chlorine	diamond	ethanol
glucose	iron(III) oxide	limestone	nitrogen	oxygen

Answer the questions using the list of substances.

Each substance may be used once, more than once or not at all.

State which of the substances:

(a) is a reactant in photosynthesis

..... [1]

(b) is the main constituent of bauxite

..... [1]

(c) are **two** products of fermentation

..... and [2]

(d) is used as a fuel

..... [1]

(e) is a gas used to convert iron into steel

..... [1]

(f) is a greenhouse gas

..... [1]

(g) is a gas that is approximately 78% of clean, dry air

..... [1]

(h) is a form of carbon.

..... [1]

[Total: 9]

- 1 The symbols of the elements of Period 3 of the Periodic Table are shown.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

Answer the following questions about these elements.

Each symbol may be used once, more than once or not at all.

Write the symbol of the element which:

- (a) forms a stable ion with a 2+ charge [1]
- (b) is the least reactive in the period [1]
- (c) is used in water treatment [1]
- (d) forms an oxide which is the main impurity in iron ore [1]
- (e) is an important component of fertilisers [1]
- (f) is stored under oil [1]
- (g) is used in food containers [1]
- (h) is found in the ore zinc blende. [1]

[Total: 8]

Metals / Properties, extraction, reactivity and corrosion of metals

Question 2 starts on the next page.

1 A list of oxides, **A** to **H**, is shown.

- A** calcium oxide
- B** aluminium oxide
- C** silicon(IV) oxide
- D** sulfur dioxide
- E** carbon dioxide
- F** iron(III) oxide
- G** silver oxide
- H** carbon monoxide

Answer the following questions about the oxides, **A** to **H**.
Each letter may be used once, more than once or not at all.

State which of the oxides, **A** to **H**:

(a) is responsible for acid rain

..... [1]

(b) has a giant covalent structure

..... [1]

(c) is a reducing agent in the blast furnace

..... [1]

(d) is the main constituent of bauxite

..... [1]

(e) is the main impurity in iron ore

..... [1]

(f) can be reduced by heating with copper.

..... [1]

[Total: 6]

3 Order of reactivity can be determined by displacement reactions.

(a) A student investigates the reactivities of four metals by carrying out a series of experiments.

Each of the metals lead, manganese, silver and zinc are added separately to aqueous metal nitrates of the other metals.

(i) Table 3.1 shows some of the results.

Table 3.1

aqueous solution	lead Pb	manganese Mn	silver Ag	zinc Zn
lead(II) nitrate		✓		
manganese(II) nitrate				
silver nitrate	✓	✓		✓
zinc nitrate	x	x		

key

✓ = displacement reaction occurs

x = displacement reaction does not occur

Complete Table 3.1 and place the four metals in their order of reactivity with the most reactive first.

1 most reactive

2

3

4

[3]

(ii) Suggest why the metal nitrates and not the metal sulfates of these four metals are used as the aqueous solutions.

..... [1]

(iii) Write the symbol equation for the reaction between zinc and silver nitrate.

..... [2]

(b) The reactivity of Group VII elements can be investigated experimentally.

A student bubbles chlorine gas into a test-tube containing aqueous potassium bromide.

(i) Describe the colour change seen in the test-tube.

from to [2]

(ii) Complete the ionic equation for this reaction.

Include state symbols.

..... +Br⁻(aq) → + [3]

(iii) The reactivity trend seen in Cl, Br and I applies to all the elements in Group VII.

Use the Periodic Table to identify the Group VII element which **cannot** displace any other Group VII elements.

..... [1]

[Total: 12]

3 Iron is manufactured in a blast furnace.

(a) Three of the starting materials added to the blast furnace are coke, iron ore and limestone.

Name the **other** starting material added to the blast furnace.

..... [1]

(b) The source of iron in the blast furnace is Fe_2O_3 . Fe_2O_3 is found in iron ore.

(i) Name the main ore of iron which contains Fe_2O_3 .

..... [1]

(ii) The iron in Fe_2O_3 is reduced by reaction with carbon monoxide. The unbalanced symbol equation is shown.

Complete the equation.



(iii) State the change in oxidation number of iron in the reaction in (ii).

from to [2]

(iv) Explain how the change of oxidation number shows that iron has been reduced.

..... [1]

(c) The major impurity in iron ore is silicon(IV) oxide. Limestone is added to the blast furnace to remove this impurity.

Write two symbol equations to show how silicon(IV) oxide is removed. For each equation, state the type of chemical reaction that takes place.

equation 1

type of chemical reaction

equation 2

type of chemical reaction

[4]

(d) Iron is converted to steel by mixing it with carbon and other elements.

(i) State the term given to a substance which is a mixture of a metal and other elements.

..... [1]

(ii) Name **one** element, other than carbon, mixed with iron in the making of stainless steel.

..... [1]

(e) Preventing the rusting of steel is important.

State the chemical name of rust.

..... [1]

(f) Steel can be coated with zinc to prevent rusting. This provides both a barrier method and sacrificial protection.

(i) State the term used for coating steel with zinc.

..... [1]

(ii) Describe another barrier method for preventing rusting.

..... [1]

(iii) Explain how zinc provides sacrificial protection.

.....

..... [2]

[Total: 17]

5 This question is about iron.

(a) (i) Describe the bonding in a metallic element such as iron.

You may include a labelled diagram as part of your answer.

.....
.....
..... [3]

(ii) Explain why iron conducts electricity when it is solid.

..... [1]

(b) Iron is extracted from hematite in the blast furnace as shown in Fig. 5.1.

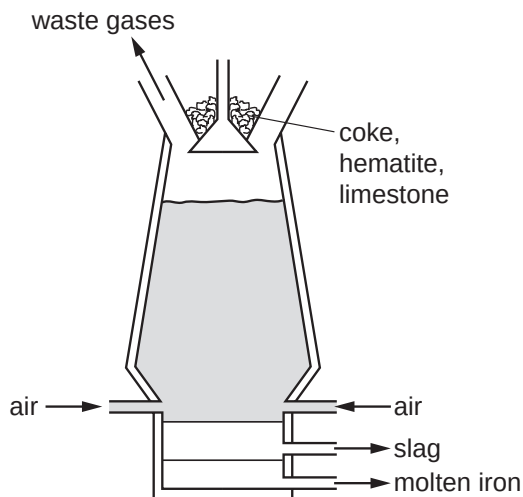


Fig. 5.1

(i) Give **two** reasons why coke is added to the blast furnace.

- 1
- 2 [2]

(ii) Explain how limestone removes the impurities in the hematite.

-
-
- [2]

(iii) Hematite contains iron(III) oxide.

Write a symbol equation for the conversion of iron(III) oxide to iron in the blast furnace.

- [2]

(iv) Suggest why the iron produced in the blast furnace is molten.

- [1]

- (c) Most iron is converted into steel. Steel is an alloy.

Steel is more useful than pure iron because it is harder and stronger.

Explain why the structure of alloys causes them to be harder and stronger than pure metals.

You may include a diagram as part of your answer.

.....
..... [2]

- (d) Iron forms rust.

Rusting is prevented by coating iron with zinc.

- (i) Name the substances that react with iron to form rust.

..... [1]

- (ii) Name the process in which zinc is used to coat iron to prevent rusting.

..... [1]

- (iii) Explain how the coating of zinc prevents rusting if the zinc is **not** scratched.

..... [1]

- (iv) When zinc is scratched the iron becomes exposed.

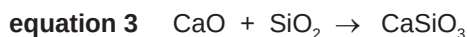
Explain how the zinc continues to prevent rusting.

.....
.....
..... [2]

[Total: 18]

- 1 Iron ore contains iron(III) oxide, Fe_2O_3 . A blast furnace is used to extract iron from Fe_2O_3 .

Equations for some of the reactions in the blast furnace are shown.



- (a) **Equation 1** shows the combustion of carbon in the blast furnace.

- (i) Name the substance which provides the carbon for this reaction.

..... [1]

- (ii) State the purpose of the combustion of carbon in the blast furnace.

..... [1]

- (b) Iron(III) oxide, Fe_2O_3 , in iron ore is converted to iron when it reacts with carbon monoxide, CO , in the blast furnace.

- (i) Calculate the percentage by mass of iron in iron(III) oxide, Fe_2O_3 .

percentage =% [2]

- (ii) State the name of the iron ore which consists mainly of iron(III) oxide.

..... [1]

- (iii) Describe how carbon monoxide is formed in the blast furnace.

..... [1]

- (iv) Write the symbol equation to show the reaction that occurs when iron(III) oxide is converted to iron in the blast furnace.

..... [2]

- (v) Name the chemical process which happens to iron when iron(III) oxide is converted to iron in the blast furnace.

..... [1]

- (c) State the type of reaction shown by **equation 2**.

..... [1]

(d) (i) Explain why the reaction in **equation 3** can be described as an acid–base reaction.

.....
.....
..... [2]

(ii) State:

- the chemical name of SiO_2

.....

- the common name given to CaSiO_3 when it is formed in the blast furnace.

..... [2]

(e) Aluminium **cannot** be extracted from its ore using a blast furnace.

(i) State why aluminium is **not** extracted from its ore using a blast furnace.

..... [1]

(ii) Name the process used to extract aluminium from its ore.

..... [1]

(f) Both iron(III) oxide and aluminium oxide contain metal ions with a 3+ charge.

(i) Write the electronic configuration of an Al^{3+} ion.

..... [1]

(ii) Deduce the number of protons and electrons in an Fe^{3+} ion.

protons	electrons

[2]

[Total: 19]

6 This question is about metals.

(a) Fig. 6.1 shows a blast furnace used to extract iron from its ore.

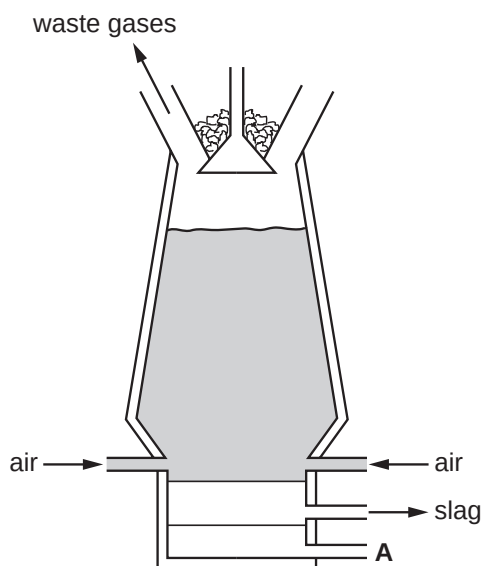


Fig. 6.1

(i) Coke and iron ore are added at the top of the blast furnace.

Name one **other** substance that is added at the top of the blast furnace.

..... [1]

(ii) Name the substance that leaves the blast furnace at **A**.

..... [1]

(iii) Slag is produced from an impurity in iron ore.

Name the impurity in iron ore that is converted into slag.

..... [1]

(iv) Name **two** substances that react together to produce the high temperature in the blast furnace.

..... and [1]

(v) Name **two** waste gases that leave the blast furnace.

1

2

[2]

(b) Zinc is produced from zinc oxide in a furnace.

The zinc is produced as a gas. It then forms molten zinc.

(i) Suggest why the zinc produced inside the furnace is a gas.

..... [1]

(ii) State the name of the physical change that occurs when gaseous zinc is converted into molten zinc.

..... [1]

(c) Zinc is used to coat iron to prevent rusting.

(i) Name the process used to coat iron with zinc as a method of rust prevention.

..... [1]

(ii) When the zinc coating is scratched, the iron underneath does not rust.

Explain why the iron underneath the zinc does **not** rust.

.....

..... [2]

(d) Zinc oxide neutralises both acids and bases.

(i) State the general name given to oxides that neutralise both acids and bases.

..... [1]

(ii) When zinc oxide neutralises aqueous sodium hydroxide, sodium zincate is formed.

The formula of the zincate ion is ZnO_2^{2-} .

Deduce the formula of sodium zincate.

..... [1]

(iii) Name the zinc compound that forms when zinc oxide neutralises dilute sulfuric acid.

..... [1]

[Total: 14]

2 A list of five metals is shown.

copper
iron
magnesium
potassium
silver

(a) All metals form positive ions.

(i) Describe how atoms form positive ions.

..... [1]

(ii) State which of the five metals in the list has the greatest tendency to form positive ions.

..... [1]

(iii) Suggest **one** of the five metals in the list which is **not** likely to show catalytic properties.

..... [1]

(iv) State which of the five metals in the list is a major component of stainless steel.

..... [1]

(b) A student adds a sample of a metal to an aqueous metal salt in a beaker to see if a displacement reaction takes place.

Complete Table 2.1 to show the colour of the solution in the beaker at the start and at the end of the experiment.

Table 2.1

metal	aqueous solution	colour at the start	colour at the end
magnesium	iron(II) sulfate	green	
silver	copper(II) sulfate		

[3]



4

(c) Most Group II metals form a gas when placed into cold water. An alkaline solution is also formed.

(i) Name the gas formed when strontium is added to cold water.

..... [1]

(ii) Name the alkaline solution formed when strontium is added to cold water.

..... [1]

(iii) One Group II metal reacts very slowly when placed in cold water. When heated, the metal reacts with steam to form a white solid.

Identify this metal and name the white solid formed.

metal

white solid [2]

(d) Under certain conditions, iron will react with steam to form an oxide of iron with the formula Fe_3O_4 .

Fe_3O_4 reacts with dilute hydrochloric acid to form a mixture of iron(II) and iron(III) salts and water.

Deduce the symbol equation for the reaction between Fe_3O_4 and dilute hydrochloric acid.

..... [3]

[Total: 14]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

6 This question is about iron.

(a) Fig. 6.1 shows a blast furnace used to extract iron from its ore.

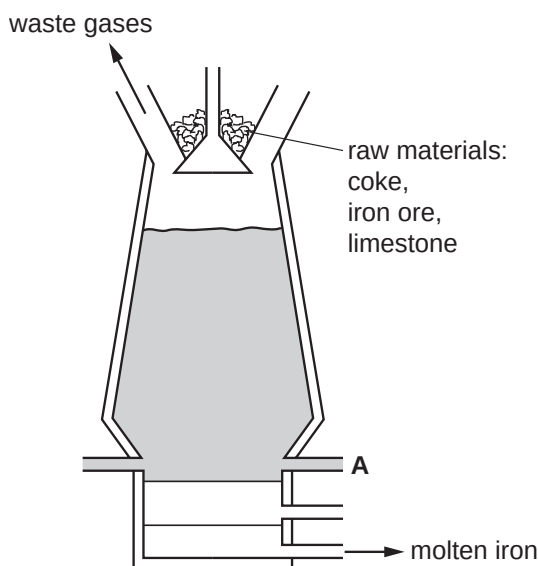


Fig. 6.1

(i) Name the main ore of iron used in the blast furnace.

..... [1]

(ii) Name the substance that enters the blast furnace at A.

..... [1]

(iii) Name the reducing agent in the extraction of iron in the blast furnace.

..... [1]

(iv) Explain why limestone is added to the blast furnace. Give details of the chemical reactions that are involved.

.....
.....
.....
..... [3]

(b) The list shows the properties of some elements.

- act as catalysts
- have low densities
- have low melting points
- form acidic or basic oxides
- form coloured compounds
- form positive or negative ions

Iron is a transition metal. Sodium is a Group I metal.

State which property from the list:

(i) is true for sodium but **not** iron

..... [1]

(ii) is true for iron but **not** sodium

..... [1]

(iii) is true for both sodium and iron

..... [1]

(iv) is **not** true for sodium and **not** true for iron.

..... [1]

(c) Steel consists mainly of iron.

Iron rusts when it reacts with water and oxygen.

Fig. 6.2 shows magnesium blocks attached to the bottom of a steel boat. The magnesium does **not** completely cover the steel.

The magnesium blocks provide sacrificial protection for the steel.

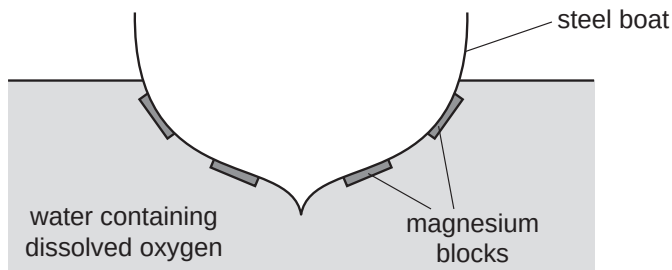


Fig. 6.2

(i) Explain, in terms of electrons, why magnesium is used for sacrificial protection.

.....
..... [2]

(ii) Name a metal that cannot provide sacrificial protection for steel.

..... [1]

[Total: 13]

1 A list of substances is shown.

bauxite
carbon dioxide
cryolite
ethane
ethanol
ethene
graphite
helium
hematite
hydrogen
silicon(IV) oxide
sodium chloride

Answer the following questions using only the substances from the list.

Each substance may be used once, more than once or not at all.

State which substance:

(a) is manufactured by fermentation

..... [1]

(b) is monatomic

..... [1]

(c) is a reactant in photosynthesis

..... [1]

(d) is a solvent in the extraction of aluminium

..... [1]

(e) is an ore of iron

..... [1]

(f) is manufactured from methane

..... [1]



DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN



(g) is a compound with a giant covalent structure

..... [1]

(h) is used as a lubricant

..... [1]

(i) is tested for with limewater.

..... [1]

[Total: 9]



3 This question is about copper and compounds of copper.

(a) (i) Describe the bonding in a metallic element such as copper.

.....
.....
.....
.....
..... [3]

(ii) Explain how solid copper conducts electricity.

..... [1]

(b) Copper is in alloys such as brass.

(i) State **one** reason why alloys are more useful than pure metals.

..... [1]

(ii) Name the substance that is present in brass, other than copper.

..... [1]

(c) Copper(II) sulfate crystals are made by the reaction between copper(II) carbonate and dilute sulfuric acid, using the following steps.

The sulfuric acid has a concentration of 0.100 mol/dm^3 .

step 1 Powdered copper(II) carbonate is added to dilute sulfuric acid. The mixture is stirred. A reaction occurs.

step 2 More copper(II) carbonate is added, with stirring, until the reaction stops.

step 3 Unreacted copper(II) carbonate is separated from aqueous copper(II) sulfate by filtration.

step 4 Aqueous copper(II) sulfate is heated until some of the water evaporates.

step 5 The remaining solution is allowed to cool and crystallise.

step 6 The crystals are removed and dried.

(i) Give **two** observations in **step 1**.

1
2
[2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN



DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN



(ii) State why the reaction stops in **step 2**.

..... [1]

(iii) Name the residue in **step 3**.

..... [1]

(iv) Name a substance, other than copper(II) carbonate, that can be added to dilute sulfuric acid to produce aqueous copper(II) sulfate.

..... [1]

(v) The solution at the end of **step 4** contains the maximum amount of copper(II) sulfate that will dissolve at that temperature.

State the term used to describe this type of solution.

..... [1]

(vi) **Step 1** is repeated using sulfuric acid of concentration 0.200 mol/dm^3 instead of 0.100 mol/dm^3 .

All other conditions are the same.

The rate of reaction increases.

Explain why the rate of reaction increases. Give your answer in terms of particles.

.....

.....

..... [2]

(vii) Copper(II) sulfate crystals have the formula $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

State the term used to describe a substance that is chemically combined with water.

..... [1]

[Total: 15]



2 Iron is extracted from iron ore in a blast furnace.

(a) The iron compound in the main iron ore used in the blast furnace is iron(III) oxide.

(i) State the name of the main iron ore used in the blast furnace.

..... [1]

(ii) State how the main source of heat is provided in the blast furnace.

..... [1]

(iii) Name the gaseous reducing agent in the blast furnace.

..... [1]

(iv) Write a symbol equation for the reduction of iron(III) oxide by the gaseous reducing agent in (a)(iii).

..... [2]

(v) The main impurity in iron ore is silicon(IV) oxide.

Write **two** symbol equations to show the role of limestone in removing silicon(IV) oxide from iron ore.

1

2 [2]

(b) Table 2.1 shows four different isotopes of iron, **A**, **B**, **C** and **D**.

Table 2.1

A	B	C	D
^{54}Fe	^{56}Fe	^{57}Fe	^{58}Fe

Explain why **A**, **B**, **C** and **D** are isotopes of iron.

.....
 [2]





(c) Iron can form two different ions, Fe^{2+} and Fe^{3+} .

- (i) Complete Table 2.2 to show the number of protons, neutrons and electrons in the two iron ions shown.

Table 2.2

	$^{54}\text{Fe}^{2+}$	$^{58}\text{Fe}^{3+}$
protons		
neutrons		
electrons		

[3]

- (ii) Describe a test for aqueous Fe^{2+} ions.

test

observations

[2]

(d) An oxidising agent converts Fe^{2+} ions to Fe^{3+} ions.

- (i) Define the term oxidising agent.

.....

..... [2]

- (ii) Explain why the conversion of Fe^{2+} ions to Fe^{3+} ions is described as oxidation.

..... [1]

- (iii) The oxidising agent for the conversion of Fe^{2+} ions to Fe^{3+} ions is aqueous potassium manganate(VII).

State **one** condition needed for the oxidation to take place.

..... [1]

(e) Iodide ions, I^- , reduce Fe^{3+} ions to Fe^{2+} ions in aqueous solution.

Suggest the identity of the other product formed in this reaction.

..... [1]

[Total: 19]

